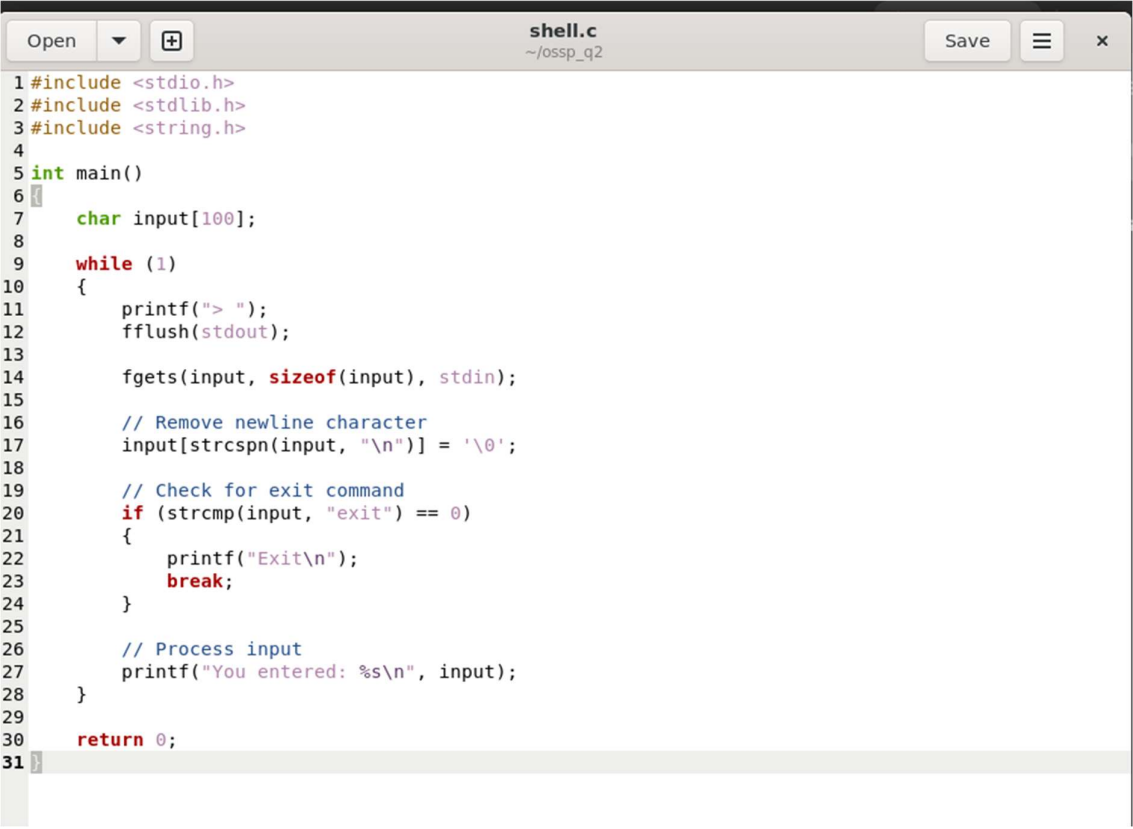


OS Skill – 2

Problem 1



```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 int main()
6 {
7     char input[100];
8
9     while (1)
10    {
11        printf("> ");
12        fflush(stdout);
13
14        fgets(input, sizeof(input), stdin);
15
16        // Remove newline character
17        input[strcspn(input, "\n")] = '\0';
18
19        // Check for exit command
20        if (strcmp(input, "exit") == 0)
21        {
22            printf("Exit\n");
23            break;
24        }
25
26        // Process input
27        printf("You entered: %s\n", input);
28    }
29
30    return 0;
31 }
```

```
siddharth@SiddharthReddy:~$ mkdir -p ~/ossp_q2
siddharth@SiddharthReddy:~$ cd ~/ossp_q2
siddharth@SiddharthReddy:~/ossp_q2$ pwd
/home/siddharth/ossp_q2
siddharth@SiddharthReddy:~/ossp_q2$ gedit shell.c
siddharth@SiddharthReddy:~/ossp_q2$ gcc shell.c -o shell
siddharth@SiddharthReddy:~/ossp_q2$ ls
shell  shell.c
siddharth@SiddharthReddy:~/ossp_q2$ ./shell
> hello
You entered: hello
> test command
You entered: test command
> exit
Exit
siddharth@SiddharthReddy:~/ossp_q2$ |
```

Problem 2

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <termios.h>
4
5 #define BUFFER_SIZE 100
6
7 void enable_raw_mode()
8 {
9     struct termios terminal;
10    tcgetattr(STDIN_FILENO, &terminal);
11    terminal.c_lflag &= ~(ICANON | ECHO);
12    tcsetattr(STDIN_FILENO, TCSANOW, &terminal);
13 }
14
15 void disable_raw_mode()
16 {
17     struct termios terminal;
18    tcgetattr(STDIN_FILENO, &terminal);
19    terminal.c_lflag |= (ICANON | ECHO);
20    tcsetattr(STDIN_FILENO, TCSANOW, &terminal);
21 }
22
23 int main()
24 {
25     char buffer[BUFFER_SIZE];
26     int index = 0;
27     char ch;
28
29     enable_raw_mode();
30
31     printf("> ");
32     fflush(stdout);
33 }
```

C ▾ Tab Width: 8 ▾ Ln 96, Col 2 INS

```
Open  keyboard.c  Save  x
~/ossp_q2

38  fflush(stdout);
39
40  while (1)
41  {
42      read(STDIN_FILENO, &ch, 1);
43
44      /* Handle Enter key */
45      if (ch == '\n' || ch == '\r')
46      {
47          buffer[index] = '\0';
48
49          printf("\nYou entered: %s\n", buffer);
50          printf("Buffer: %s\n", buffer);
51
52          /* Handle exit command */
53          if (index == 4 &&
54              buffer[0] == 'e' &&
55              buffer[1] == 'x' &&
56              buffer[2] == 'i' &&
57              buffer[3] == 't')
58          {
59              printf("Exit\n");
60              break;
61          }
62
63          /* Reset input buffer */
64          index = 0;
65
66          printf("> ");
67          fflush(stdout);
68      }
69
70      /* Handle Backspace */
71      else if (ch == 127 || ch == '\b')
72      {
73          if (index > 0)
74          {
```

C ▾ Tab Width: 8 ▾ Ln 96, Col 2 INS

```
70      /* Handle Backspace */
71      else if (ch == 127 || ch == '\b')
72      {
73          if (index > 0)
74          {
75              index--;
76
77              printf("\b \b");
78              fflush(stdout);
79          }
80      }
81
82      /* Handle normal keyboard characters */
83      else if (index < BUFFER_SIZE - 1)
84      {
85          buffer[index] = ch;
86          index++;
87
88          printf("%c", ch);
89          fflush(stdout);
90      }
91  }
92
93  disable_raw_mode();
94
95  return 0;
96 }
```

C ▾ Tab Width: 8 ▾ Ln 96, Col 2 INS

```
siddharth@SiddharthReddy:~/ossp_q2$ gedit shell.c
siddharth@SiddharthReddy:~/ossp_q2$ gedit keyboard.c
siddharth@SiddharthReddy:~/ossp_q2$ gcc keyboard.c -o keyboard
siddharth@SiddharthReddy:~/ossp_q2$ ./keyboard
> hello
You entered: hello
> hello
You entered: hello
> test command
You entered: test command
> exit
You entered: exit
Exit
siddharth@SiddharthReddy:~/ossp_q2$ gedit keyboard.c
siddharth@SiddharthReddy:~/ossp_q2$ gcc keyboard.c -o keyboard
siddharth@SiddharthReddy:~/ossp_q2$ ./keyboard
> hello
You entered: hello
Buffer: hello
> hello
You entered: hello
Buffer: hello
> hello
You entered: hello
Buffer: hello
> |
```